

Case Study - 1

Green Optimization on EdTech Platform's Video Transcoding

CSE407 [Green Computing]

Section: 3 [Summer 2026]

Group: 07

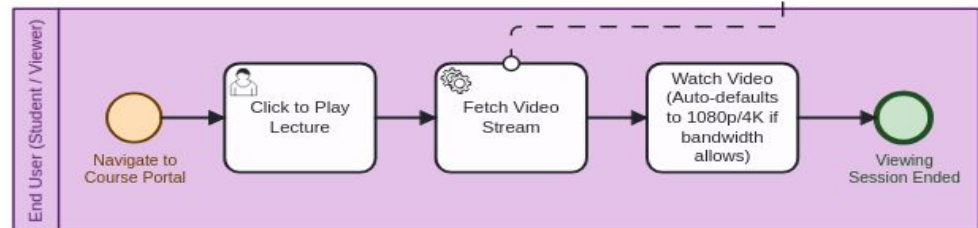
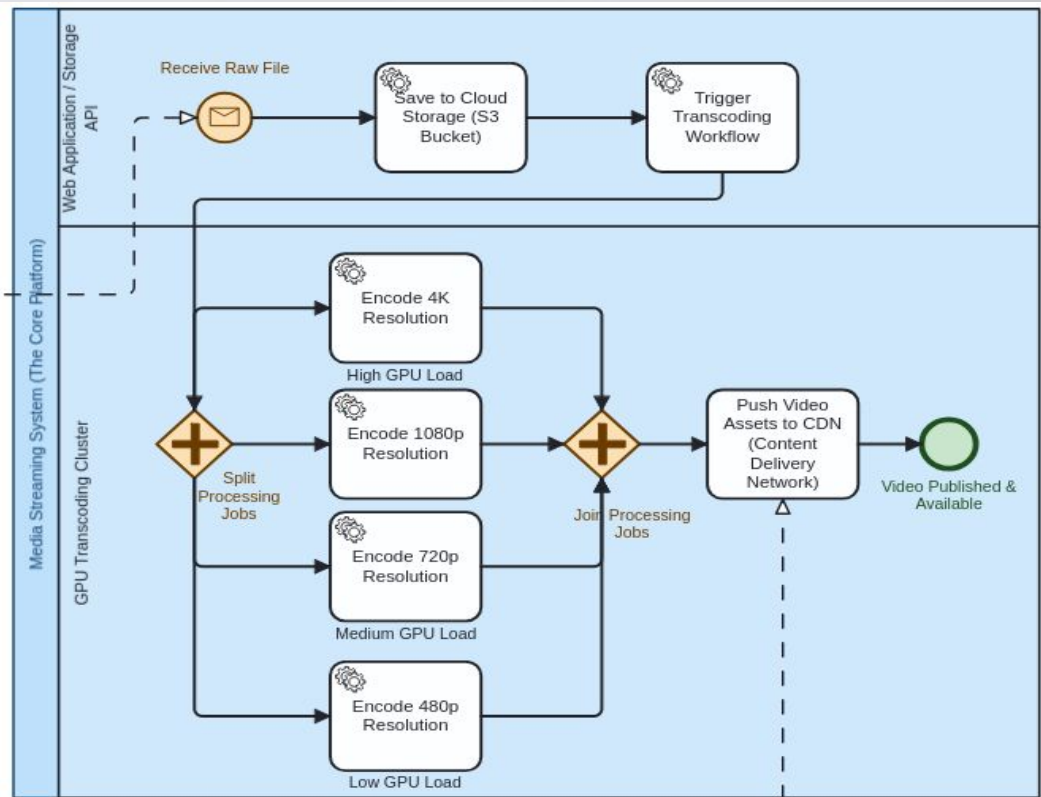
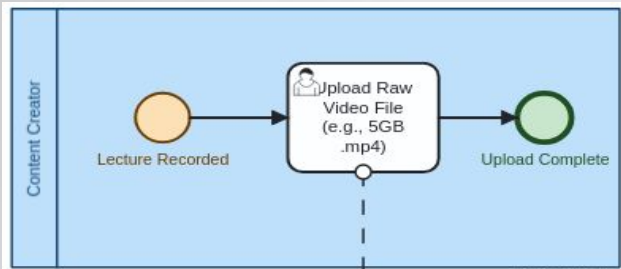
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Introduction and Objectives

Educational media platforms process massive amounts of video data daily. The traditional architectures powering these platforms prioritize speed over sustainability, leading to massive compute waste and high carbon emissions.

- **Objective:** To model, analyze, and optimize a legacy video transcoding pipeline using Business Process Model and Notation (BPMN), reducing absolute energy consumption and carbon footprint without disrupting the core educational service.
- **Reference Architectures:** Our baseline models are derived from the standard video ingestion frameworks utilized by leading EdTech platforms, including:
 - Open edX (openedx.org)
 - Khan Academy (khanacademy.org)





The AS-IS Process: Current Pipeline Mechanics

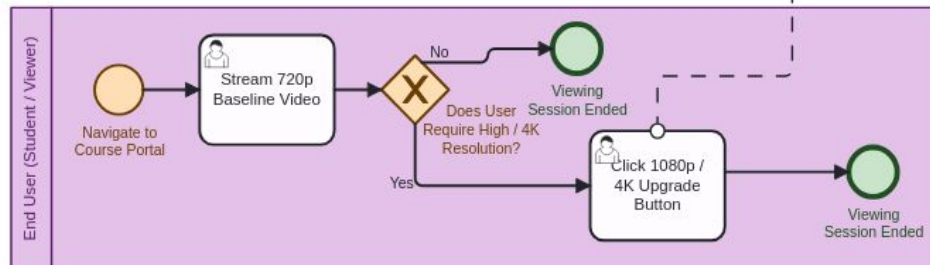
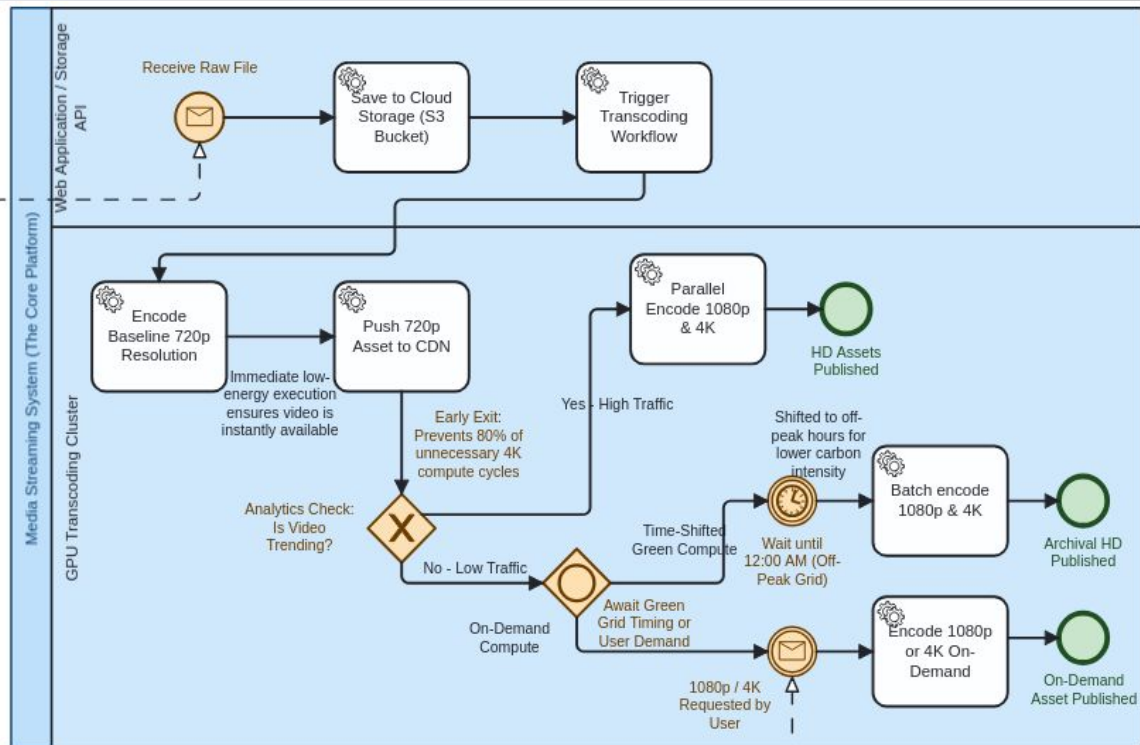
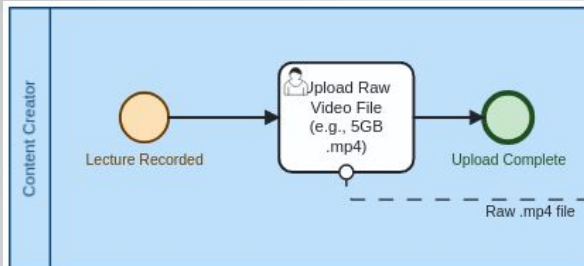
- **Trigger:** An educator uploads a raw lecture file (often large, uncompressed formats) to the platform's content management system.
- **Immediate Processing:** The upload instantly triggers the cloud compute cluster to begin transcoding.
- **Parallel Execution:** The system forces the GPU cluster to encode the video into every possible resolution (4K, 1080p, 720p, 480p) concurrently.
- **Blind Delivery:** The system pushes all high-definition assets to the Content Delivery Network (CDN) simultaneously, regardless of whether students have the bandwidth or desire to stream in 4K.

Phase 2: Identifying Optimization Opportunities (Part 1)

- **Compute Hotspot:** The system over-processes data. Energy is expended encoding 4K and 1080p resolutions for 100% of uploaded videos.
- **The Reality of Demand:** Analytics show that a significant majority of educational videos are watched in standard definition (720p), either due to mobile viewing or bandwidth constraints.
- **Optimization Opportunity:** Implement Decision Point Analysis. By adding a logic gateway to check if a video is actively trending or explicitly requested in 4K, the system can bypass the heaviest GPU workloads for the majority of the catalog.

Phase 2: Identifying Optimization Opportunities (Part 2)

- **Carbon Timing Hotspot:** Because the AS-IS process triggers immediately upon upload, the heaviest processing typically occurs during standard daytime working hours.
- **Grid Carbon Intensity:** Daytime electrical grids often rely on high-carbon fossil fuel generation to meet peak demand.
- **Optimization Opportunity:** Implement Time-Shifting. By utilizing BPMN timer events, non-urgent batch processing (like archival 4K generation) can be delayed until off-peak nighttime hours, taking advantage of lower grid demand and higher renewable energy availability.



Phase 3: Energy and Carbon Calculations

Assumptions: 1,000 videos/month. GPU Power: 0.3 kW. Daytime Grid: 400 g CO2eq/kWh. Nighttime Grid: 150 g CO2eq/kWh.

Metric	AS-IS (Current)	TO-BE (Optimized)
Daytime Energy (kWh)	1,800 kWh	600 kWh (720p Baseline only)
Nighttime Energy (kWh)	0 kWh	240 kWh (1080p/4K Upgrades)
Total Monthly Energy	1,800 kWh	840 kWh
Day time Carbon (g CO2eq)	720,000 g	240,000 g
Night time Carbon (g CO2eq)	0 g	36,000 g
Total Carbon Footprint	720,000 g CO2eq	276,000 g CO2eq

Validation

- **The Efficiency Gain:** The new BPMN architecture successfully cuts monthly hardware energy use by 53.3% (960 kWh saved) and drops the total carbon footprint by 61.7% (444,000 g CO₂eq avoided).
- **System Validation:** We achieved this purely through workflow orchestration, without needing to rewrite any core transcoding algorithms.

Trade-offs

- **Availability:** Videos that are not highly viewed will only be available in 720p instantly; 4K versions will require an overnight wait or a slight on-demand loading buffer.
- **Complexity:** The system requires a more robust internal analytics engine to monitor "trending" status to trigger the correct BPMN gateways.

References

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